Solutions: Problem Set 3

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Problem 3.1 (Domain estimation). We use the data set `sepe.dat` from Problem Set 1. Treat the data as a simple random sample without replacement (SRS) of $n = 800$ establishments from a population of size $N = 3770$. We focus on industry class 31, that is, the domain U_{31} identified by `code == 1`.

a) Assume that the domain size N_{31} is unknown. Estimate the domain population mean $\bar{y}_{U31}$ of environmental-protection expenditures and its standard error.

> **Hint.** If `dn` is a survey design created with `svydesign()`, use `subset(dn, code == 1)` to restrict the design to the domain. Do not create a new SRS design from the domain observations: under the original SRS, the domain sample size n_{31} is random.

Solution. First load the data set `sepe.dat`, which is an SRS of size $n = 800$, and define the population size $N = 3770$:

```
> sepe <- read.table("http://stat.ethz.ch/Teaching/Datasets/WBL/sepe.dat",
                     header = TRUE)
> sepe$I31 <- as.integer(sepe$code == 1)
> N <- 3770
```

Calculation with the `survey` package: First define the survey design object for the full sample:

```
> library("survey")
> sepe$fpc <- N
> dn_sepe <- svydesign(ids = ~1, fpc = ~fpc, data = sepe)
```

For domain estimation, it is important to distinguish between subsetting a survey design object and creating a new design object from the observations in the domain. The command

```
> dn_sepe_31 <- subset(dn_sepe, I31 == 1)
```

retains the original sampling design. Thus, establishments in industry class 31 are treated as a domain of the original SRS of $n = 800$ establishments. In particular, the number n_{31} of sampled domain units is random.

The domain population mean $\bar{y}_{U31}$ and its standard error are estimated by

```
> m_y_31 <- svymean(~expend, dn_sepe_31)
> m_y_31
```

```
          mean     SE
expend 113052 6432.8
```

By contrast, suppose that we first restrict the data to industry class 31 and then define a new design:

```
> sepe_31 <- subset(sepe, I31 == 1)
> dn_sepe_31_new <- svydesign(ids = ~1, data = sepe_31)
> svymean(~expend, dn_sepe_31_new)

          mean     SE
expend 113052 7269.4
```

This specifies a different sampling design. It treats the n_{31} observed establishments as if they formed a new equal-probability sample, rather than a domain of the original SRS. The estimated mean is the same because both estimators reduce to the sample mean within the domain. The estimated standard errors differ, however, because the two design objects represent different sampling designs.

Remarks. Let $\mathbb{1}_{i,31} = \mathbb{1}\{i \in U_{31}\}$ denote the domain indicator and let $s_{31} = s \cap U_{31}$. Then

$$n_{31} = \sum_{i \in s} \mathbb{1}_{i,31}.$$

The domain mean is estimated by

$$\bar{y}_{31} = \frac{\sum_{i \in s} \mathbb{1}_{i,31}\, y_i}{\sum_{i \in s} \mathbb{1}_{i,31}} = \frac{1}{n_{31}} \sum_{i \in s_{31}} y_i.$$

Because n_{31} is random under the original SRS, this estimator can be viewed as a ratio estimator. A linearization variance estimator under SRS is

$$\widehat{\mathsf{var}}\,(\bar{y}_{31}) = (1-f)\frac{n(n_{31}-1)}{n_{31}^2(n-1)}\,\widehat{\sigma}^2_{y,31}, \qquad f = \frac{n}{N},$$

where $\widehat{\sigma}^2_{y,31}$ is the sample variance of y among the n_{31} sampled domain units.

Calculation with R:

```
> n31 <- sum(sepe$I31)
> mean_y_31 <- mean(sepe$expend[sepe$I31 == 1])
> sigma2_y_31 <- var(sepe$expend[sepe$I31 == 1])
> se_mean_y_31 <- sqrt((1 - n / N) * n * (n31 - 1) /
                       (n31^2 * (n - 1)) * sigma2_y_31)
```

The manually calculated quantities agree with those obtained from the subsetted survey design.

```
> identical(as.numeric(coef(m_y_31)), mean_y_31)

[1] TRUE

> all.equal(as.numeric(SE(m_y_31)), se_mean_y_31)

[1] TRUE
```

b) Now assume that the domain size $N_{31} = 650$ is known, but that this information was not used when the sample was drawn. Estimate the population domain total of environmental expenditures $t_{y,31}$ and its standard error.

Solution. The sampling design remains an SRS from the full population. Hence, n_{31} remains random. Following the classical definition of the hypergeometric distribution,[1] consider a population of N units, of which N_{31} belong to domain U_{31} and $N - N_{31}$ do not. If an SRS of n units is drawn without replacement, then the number n_{31} of sampled domain units follows a hypergeometric distribution:

$$n_{31} \sim \text{Hypergeometric}(N, N_{31}, n). \tag{1}$$

The known domain size $N_{31} = 650$ is used only at the estimation stage. The domain total is estimated by

$$\widehat{t}_{y,31} = N_{31} \cdot \bar{y}_{31},$$

with estimated standard error

$$\widehat{\mathsf{SE}}(\widehat{t}_{y,31}) = N_{31} \cdot \widehat{\mathsf{SE}}(\bar{y}_{31}).$$

The variance estimator for $\bar{y}_{31}$, and hence $\widehat{\mathsf{SE}}(\bar{y}_{31})$, is the linearization variance estimator under SRS introduced in part a).

Calculation with the `survey` package: We use the subsetted survey design `dn_sepe_31` from part a). The estimated domain mean $\bar{y}_{31}$ has already been calculated; it is reproduced here for ease of reference:

```
> m_y_31 <- svymean(~expend, dn_sepe_31)
```

The estimated domain total and its standard error are then

```
> N31 <- 650
> c(total = N31 * coef(m_y_31), SE = N31 * SE(m_y_31))
```

```
total.expend           SE
    73483508      4181303
```

> **Remarks.** If the domain size N_{31} is known, then n_{31} has a hypergeometric distribution; see (1). In our case, n_{31} has expected value ≈ 137.9 and standard deviation ≈ 9.6. Thus, the observed value $n_{31} = 138$ is close to its expected value under the sampling design. For example, the central 80% range of the distribution can be obtained with:
>
> ```
> > qhyper(c(0.1, 0.9), N31, N - N31, n)
> ```
>
> ```
> [1] 126 150
> ```
>
> Note that R uses an equivalent but different parametrization from that in (1).
>
> Another remark is in order. The known domain size N_{31} can also be incorporated by calibration. We can define the domain indicator as an auxiliary variable and calibrate the full design to the known totals N and N_{31}:
>
> ```
> > dn_sepe_cal <- calibrate(dn_sepe, ~I31,
>                            population = c(`(Intercept)` = N, I31 = N31))
> > svytotal(~expend, subset(dn_sepe_cal, I31 == 1))
> ```

[1] See e.g., Johnson, N. L, Kemp, A. W., and Kotz, S. (2005). Univariate Discrete Distributions, 3rd ed., Hoboken (NJ): John Wiley & Sons, p. 251.

```
            total      SE
expend 73483508 4181303
```

We will study calibration in more detail later.

Problem 3.2 (Planning a stratified sample). We *plan* a survey to estimate the population mean of environmental expenditures (`expend`). We want to assess whether stratification by industry group (`code`) can improve precision relative to a simple random sample without replacement (SRS).

We use the data set `sepe.dat` from Problem Set 1. Treat the data as an SRS of $n = 800$ establishments from a population of size $N = 3770$. We focus on industry class 31, that is, the domain U_{31} of size $N_{31} = 650$, identified with `code == 1`. Assume that:

- the planned net sample size is $n = 800$ establishments;
- the anticipated response rate is 80 % in both strata; and
- the within-stratum variances of `empl` and `expend` can be estimated from the existing SRS `sepe.dat`.

a) Allocate the net sample *proportionally* to the two strata. Determine the required gross sample sizes and estimate the variance of the stratified estimator of the population mean of expenditures.

Solution. The population sizes are $N_{rest} = 3120$ for `code == 0` and $N_{31} = 650$ for `code = =1`, so $N = 3770$.

```
> sepe <- read.table("http://stat.ethz.ch/Teaching/Datasets/WBL/sepe.dat",
                     header = TRUE)

> Nh <- c("rest" = 3120, "31" = 650)
> N <- sum(Nh)
> Wh <- Nh / N
> Wh

     rest        31
0.8275862 0.1724138
```

Under proportional allocation,

$$n_h \approx n\, W_h, \qquad W_h = \frac{N_h}{N}.$$

We round the stratum sample sizes while keeping their sum equal to $n = 800$.

```
> n <- 800
> n_prop <- round(n * Wh)
> n_prop["rest"] <- n - n_prop["31"]
> n_prop

rest   31
 662  138
```

Because the anticipated response rate is the same in both strata, the gross sample can also be allocated proportionally. The total gross sample size is $800/0.8 = 1000$.

```
> response_rate <- 0.8
> n_gross <- round(n / response_rate)
> n_prop_gross <- round(n_gross * Wh)
> n_prop_gross["rest"] <- n_gross - n_prop_gross["31"]
> n_prop_gross
```

```
rest   31
 828  172
```

Thus, the required gross sample sizes are 828 establishments with `code == 0` and 172 with `code == 1`.

The variance of the stratified mean $\bar{y}^{\text{strat}} = \sum_{h=1}^{H} W_h \, \bar{y}_{h,U}$ is

$$\text{var}(\bar{y}^{\text{strat}}) = \sum_{h=1}^{H} W_h^2 \left(1 - \frac{n_h}{N_h}\right) \frac{\sigma_h^2}{n_h}.$$

We estimate σ_h^2 using the existing simple random sample.

```
> sigma2_y_h <- tapply(sepe$expend, sepe$code, var)
> var_prop <- sum(Wh^2 * (1 - n_prop / Nh) * sigma2_y_h / n_prop)
> var_prop
```

```
[1] 4423859
```

The estimated variance under proportional allocation is therefore 4'423'859.

> **Remarks.** The response rate affects the *gross* sample size, but the variance calculation is based on the planned net sample sizes. Using the general variance formula also avoids any approximation caused by rounding the proportional allocation.

b) Now use a Neyman-type (optimum) allocation, but base the allocation on the within-stratum standard deviations of the auxiliary variable `empl` rather than on `expend`.

- Determine the required gross sample sizes.
- Under what conditions is it reasonable to derive the allocation from an auxiliary variable?
- Estimate the variance of the stratified estimator of mean environmental expenditures under this allocation.

Solution. The Neyman allocation for a target variable y is

$$n_h = n \, \frac{W_h \, \sigma_{y,h}^2}{\sum_{j=1}^{H} W_j \, \sigma_{y,j}^2}, \qquad h = 1, \ldots, H.$$

Here we deliberately replace $\sigma_{y,j}^2$ by the (estimates of) within-stratum standard deviations of `empl`.

```
> sigma_x_h <- tapply(sepe$empl, sepe$code, sd)
> names(sigma_x_h) <- c("rest", "31")
> alloc_x <- Wh * sigma_x_h
> n_aux <- round(n * alloc_x / sum(alloc_x))
> n_aux

rest   31
 629  171

> n_aux["rest"] <- n - n_aux["31"]
> n_aux

rest   31
 629  171
```

The corresponding gross allocation is

```
> n_aux_gross <- round(n_gross * alloc_x / sum(alloc_x))
> n_aux_gross["rest"] <- n_gross - n_aux_gross["31"]
> n_aux_gross

rest   31
 787  213
```

An auxiliary-variable allocation is useful when the pattern of within-stratum variability of the auxiliary variable resembles that of the target variable. More precisely, we would like $\sigma_{y,h} \approx c \cdot \sigma_{x,h}$ with approximately the same positive constant c across strata. In that case, the allocation based on x is close to the Neyman allocation for y. Historical values of the target variable are a common example of a useful auxiliary variable.

The existing sample can be used to inspect this condition directly:

```
> sigma_y_h <- sqrt(sigma2_y_h)
> cbind(sigma_x_h = sigma_x_h, sigma_y_h = sigma_y_h,
        ratio = sigma_y_h / sigma_x_h)

     sigma_x_h sigma_y_h    ratio
rest   177.936  62525.73 351.3945
31     231.377  85396.08 369.0776
```

The allocation was computed from `empl`, so its variance must be evaluated using the *actual* stratum sample sizes above and the within-stratum variances of `expend`:

```
> var_aux <- sum(Wh^2 * (1 - n_aux / Nh) * sigma2_y_h / n_aux)
> var_aux

[1] 4332910
```

The estimated variance is therefore 4'332'910.

c) Compare the estimated variances from parts a) and b) with the estimated variance under a simple random sample of $n = 800$. Does stratification improve precision?

Solution. Let $\bar{y}_{\text{SRS}}$ denote the sample mean for SRS. It holds that

$$\widehat{\mathsf{var}}(\bar{y}_{\text{SRS}}) = \left(1 - \frac{n}{N}\right) \frac{\sigma_y^2}{n},$$

where σ_y^2 is estimated from `sepe.dat`.

```
> var_srs <- (1 - n / N) * var(sepe$expend) / n
> var_srs
```

```
[1] 4567134
```

The three planning variances are:

```
> c(SRS = var_srs, proportional = var_prop, auxiliary = var_aux)
```

```
         SRS proportional    auxiliary
     4567134      4423859      4332910
```

Relative to simple random sampling, the estimated variance reductions are

```
> 100 * c(proportional = 1 - var_prop / var_srs,
          auxiliary = 1 - var_aux / var_srs)
```

```
proportional    auxiliary
    3.137084     5.128461
```

percent. The auxiliary-variable allocation improves on proportional allocation if $\widehat{\mathsf{var}}_{\text{aux}} < \widehat{\mathsf{var}}_{\text{prop}}$. Unlike a true Neyman allocation based on `expend`, this ordering is not guaranteed because the allocation in part b) is based on `empl`.

Problem 3.3 (PPS Sampling and the Horvitz–Thompson estimator). We use the data set `sepe.dat` from Problem Set 1 and treat the $N = 800$ establishments as the finite population U. We want to draw a PPS sample without replacement of size $n = 200$, using the number of full-time employees as the measure of size. The data set contains:

`empl`: number of full-time employees, denoted by x_i. These values are known from a register before sampling;

`expend`: environmental expenditures, denoted by y_i. We treat these values as unknown until an establishment is sampled.

a) Compute the first-order inclusion probabilities and draw a PPS sample of $n = 200$ establishments without replacement using Brewer's method. If there were no certainty units, the inclusion probabilities would be

$$\pi_i = \frac{n x_i}{t_x}, \qquad t_x = \sum_{i \in U} x_i.$$

Compare these naive probabilities with the probabilities returned by `pps_probabilities()` of the `robsurvey` package. What do you observe?

> **Hint.** Attach the packages `survey` and `robsurvey` to the search path. Compute the inclusion probabilities and draw the sample as follows:

```
pps_setup <- pps_probabilities(size = sepe$empl, n = 200)
pi <- as.numeric(pps_setup)

set.seed(11)
s <- pps_draw(pps_setup, method = "brewer")
```

Solution. We attach the packages to the search path and load the data:

```
> library("survey")
> library("robsurvey")
> sepe <- read.table("http://stat.ethz.ch/Teaching/Datasets/WBL/sepe.dat",
                     header = TRUE)
```

The valid first-order inclusion probabilities are

```
> pps_setup <- pps_probabilities(sepe$empl, n = 200)
> sepe$pi <- as.numeric(pps_setup)
> summary(sepe$pi)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.07253 0.18082 0.25000 0.36296 1.00000
```

```
> sum(sepe$pi == 1)
```

```
[1] 16
```

Hence, 16 establishments are certainty units with $\pi_i = 1$. By contrast, the naive formula gives

```
> pi_naive <- 200 * sepe$empl / sum(sepe$empl)
> summary(pi_naive)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.07102 0.17706 0.25000 0.35543 1.55791
```

```
> sum(pi_naive > 1)
```

```
[1] 15
```

Some of these values exceed one and therefore cannot be inclusion probabilities. The function `pps_probabilities()` handles this iteratively: certainty units are assigned $\pi_i = 1$, and the probabilities of the remaining units are rescaled so that

$$0 < \pi_i \leq 1 \qquad \text{and} \qquad \sum_{i \in U} \pi_i = n.$$

Thus, with certainty units present, the final probabilities are not simply nx_i/t_x for every unit. We then draw the PPS sample:

```
> set.seed(11)
> s <- pps_draw(pps_setup, method = "brewer")
> length(s)
```

```
[1] 200
```

b) Create a survey design object for the sample from part a). Then use `svyplot()` to plot environmental expenditures against the number of employees. The bubble sizes represent the sampling weights $w_i = 1/\pi_i$. Describe the main features of the plot.

Remarks. For the PPS design of part a), use

```
dn_sepe <- svydesign(ids = ~1, fpc = ~pi, pps = "brewer",
                     data = sepe[s, ])
```

Here, `fpc = ~pi` supplies the first-order inclusion probabilities. The option `pps = "brewer"` tells `survey` to use Brewer's approximation for the second-order inclusion probabilities needed for variance estimation. This is distinct from Brewer's sampling algorithm used in part a).

Solution. We define the design by

```
> dn_sepe <- svydesign(ids = ~1, fpc = ~pi, pps = "brewer",
                       data = sepe[s, ])
```

Omitting `fpc` would instead create a with-replacement approximation to the design.

The scatterplot (Figure 1) is obtained with the commands:

```
> svyplot(expend ~ empl, dn_sepe, inches = 0.1,
          xlab = "Number of employees",
          ylab = "Environmental expenditures")
> certainty_cut <- min(sepe$empl[sepe$pi == 1])
> abline(v = certainty_cut, lty = 2)
```

The main features are:

- Establishments with fewer employees have smaller inclusion probabilities and therefore larger sampling weights. They appear as larger bubbles.
- The largest establishments are certainty units. The dashed line marks the smallest value of `empl` among these units.
- The size measure `empl` is positively related to `expend`: 0.57. PPS sampling is particularly effective when y_i is closely related to the size measure, so that the ratios y_i/x_i do not vary too much.

c) Estimate the population total of environmental expenditures,

$$t_y = \sum_{i \in U} y_i,$$

using the Horvitz–Thompson estimator and the design from part b). Also compute an estimate of the standard error of the estimator.

Solution. The Horvitz–Thompson estimate and its estimated standard error are obtained directly with

```
> t_y <- svytotal(~expend, dn_sepe)
> t_y
```

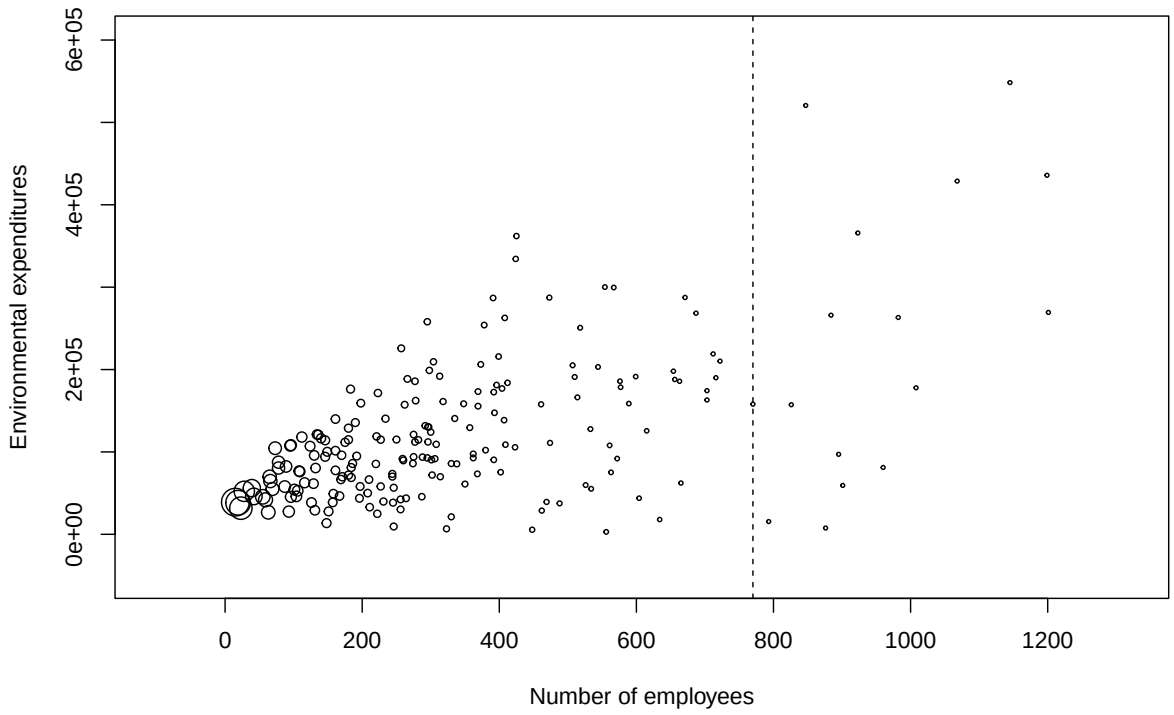


Figure 1: Environmental expenditures vs. number of employees; the size of the bubbles is proportional to the sampling weights

```
          total      SE
expend 71293708 3300926
```

The PPS weights have an additional useful property. Because the inclusion probabilities were constructed from `empl`, the Horvitz–Thompson estimator reproduces its known population total:

```
> t_x <- svytotal(~empl, dn_sepe)
> c(estimate = coef(t_x), population = sum(sepe$empl))
```

```
estimate.empl    population
       154181        154181
```

The point estimate is therefore exact, but the default variance estimator does not automatically use this calibration information. We can tell `survey` that the supplied weights are already calibrated to t_x:

```
> dn_sepe_calib <- svydesign(ids = ~1, fpc = ~pi, pps = "brewer",
                             data = sepe[s, ],
                             calibrate.formula = ~ -1 + empl)
> svytotal(~empl, dn_sepe_calib)
```

```
      total SE
empl 154181  0
```

The formula has no intercept because the PPS weights reproduce t_x, but they do not generally reproduce the population size N. The option `calibrate.formula` does not change the weights; it tells the variance estimator about an existing calibration equation.

The estimate of t_y is unchanged, while its estimated standard error can decrease:

```
> t_y_calib <- svytotal(~expend, dn_sepe_calib)
> c(estimate = unname(coef(t_y_calib)), SE_without_calibration = SE(t_y),
    SE_with_calibration = SE(t_y_calib))
```

```
            estimate SE_without_calibration    SE_with_calibration
            71293708               3300926                3313698
```